

Key-value Database

 systemdesignschool.io/fundamentals/key-value-database

☐ include secondary database models

68 systems in ranking, March 2023

Rank			DBMS	Database Model	Score		
Mar 2023	Feb 2023	Mar 2022			Mar 2023	Feb 2023	Mar 2022
1.	1.	1.	Redis 	Key-value, Multi-model 	172.45	-1.39	-4.31
2.	2.	2.	Amazon DynamoDB 	Multi-model 	80.77	+1.08	-1.03
3.	3.	3.	Microsoft Azure Cosmos DB 	Multi-model 	36.10	-0.40	-4.79
4.	4.	4.	Memcached	Key-value	22.61	-0.56	-3.06
5.	5.	 6.	Hazelcast	Key-value, Multi-model 	8.63	-0.48	-1.43
6.	6.	 5.	etcd	Key-value	8.60	+0.09	-3.23
7.	7.	 10.	Aerospike 	Multi-model 	6.54	-0.02	+0.31
8.	8.	 9.	Ehcache	Key-value	6.03	-0.01	-0.52
9.	9.	 14.	Google Cloud Bigtable	Multi-model 	5.44	-0.47	+1.10
10.	10.	 8.	Ignite	Multi-model 	5.29	-0.21	-1.58
11.	 12.	 7.	Riak KV	Key-value	5.28	+0.14	-1.75
12.	 11.	 11.	ArangoDB 	Multi-model 	5.04	-0.26	-0.57
13.	 15.	 15.	RocksDB 	Key-value	4.57	+0.11	+0.51
14.	 13.	 13.	OrientDB	Multi-model 	4.30	-0.24	-0.63
15.	 14.		GemFire	Key-value, Multi-model 	4.26	-0.26	
16.	 17.	 12.	Oracle NoSQL	Multi-model 	3.72	+0.02	-1.22
17.	 18.	 18.	LevelDB	Key-value	3.54	-0.01	+0.34
18.	 16.	 16.	Infinispan	Key-value	3.52	-0.24	+0.11
19.	19.	19.	Oracle Berkeley DB	Multi-model 	3.38	+0.14	+0.40
20.	 23.	 22.	InterSystems IRIS	Multi-model 	3.06	+0.25	+0.86
21.	21.	21.	LMDB	Key-value	2.99	+0.13	+0.44
22.	 20.	 17.	InterSystems Caché	Multi-model 	2.96	-0.05	-0.35
23.	 24.	23.	Amazon SimpleDB	Key-value	2.83	+0.08	+0.67

NoSQL databases optimize for different access patterns. The simplest pattern is direct key-value lookup. Many applications need only this operation: session storage, caching, user preferences.

Why Key-Value Databases Exist

In the original [Dynamo paper](#), Amazon engineers identified a pattern:

Many services on Amazon's platform only need primary-key access to a data store. Services providing best seller lists, shopping carts, customer preferences, session management, sales rank, and product catalog use relational databases inefficiently. This pattern limits scale and availability. Dynamo provides a simple primary-key only interface to meet these requirements.

Key-value databases are the simplest NoSQL type. The interface resembles a hashmap. Store values by key. Retrieve values by key. That's it.

How Key-Value Databases Work

The data model couldn't be simpler. Each record has two parts: a unique key and a value. The database provides three operations: `get(key)`, `put(key, value)`, and `delete(key)`. No schemas. No relationships. No joins.

This simplicity enables performance. **Primary key lookups are fast.** The database can hash the key to find the exact location. No table scans. No index traversals. Typical latency measures in microseconds.

Horizontal scaling is straightforward. Partition data by key ranges or hash values. Add more servers to increase capacity. Each server handles a subset of keys. The partitioning logic remains simple because queries never span partitions.

The **API matches programming language dictionaries.** Developers already understand `get` and `put`. No SQL to learn. No ORM complexity. Application code and database model align naturally.

When to Use Key-Value Databases

High-performance reads and writes benefit from key-value simplicity. Reading and writing with a key is a fast operation. Applications demanding high-speed operations choose key-value stores.

Simple data structures fit naturally. Customer details, webpages keyed by URL, shopping-cart contents, product categories, and e-commerce product details work well. Each entity needs independent access without relationships.

Basic query requirements match key-value capabilities. Applications that don't need complex queries or frequent updates across multiple records work well. Simple data storage and retrieval operations are ideal.

Popular Key-Value Databases

Because of their performance and simplicity, key-value databases rank highly in popularity according to [db engine ranking](#).

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Redis and DynamoDB lead the category. Redis specializes in in-memory caching with microsecond latency. DynamoDB provides durable, distributed storage with single-digit millisecond latency.

Real-World Examples

[Dropbox stores file metadata](#) in DynamoDB through their Alki system. Each file's metadata maps to a unique file ID. No relationships between files matter. Simple key-value lookups provide the speed needed. This saved Dropbox millions in infrastructure costs compared to their previous on-premises storage.

[Snapchat stores Stories](#) in DynamoDB. Story ID maps to content metadata. Stories don't relate to each other. Ephemeral nature matches key-value simplicity. Snap handles over 10 million queries per second and reduced median latency by 20%.

[Disney+ uses DynamoDB](#) for content metadata and user interactions. Video ID maps to title, description, and thumbnail URL. Each video loads independently. They ingest billions of customer actions daily for features like watchlists and personalized recommendations.